

CloudScale DevOps: CI/CD Pipeline & Deployment Governance

Technical Policy (POL-502) | Release Architecture & Kubernetes Standards | Effective: 2026

1. Production Deployment Windows & Freeze Policies

- **Standard Deployment Windows:** Production releases are permitted exclusively on **Tuesdays and Thursdays between 10:00 UTC and 14:00 UTC**.
- **Deployment Freeze:** Releases are strictly forbidden on Fridays after 12:00 UTC, over weekends, and during the final 5 business days of each fiscal quarter.
- **Emergency Hotfixes:** Permitted outside standard windows only with written sign-off from both the Principal DevOps Architect and the Director of Engineering.

2. Progressive Canary Rollout Schedule

- **Stage 1 (Canary Initial - 5% Traffic):** 30-minute observation window. Automated gate checks require HTTP 5xx error rate < 0.05% and p99 latency < 250ms.
- **Stage 2 (Canary Mid - 25% Traffic):** 60-minute observation window. Automated gate checks enforce memory leak detection and error rate delta < 0.1%.
- **Stage 3 (Full Promotion - 100% Traffic):** Continuous monitoring. Enforces cluster health 100% and Prometheus alert silence check.

3. Automated Rollback Criteria

The automated deployment pipeline triggers an immediate automatic rollback to the previous stable Helm release if any of the following occur:

- Error rate exceeds **0.5%** of total ingress requests over any consecutive 3-minute window.
- The 99th percentile (p99) response latency exceeds **350 milliseconds**.
- Container restart count exceeds **3 restarts per pod** within a 10-minute window (CrashLoopBackOff).

4. Production Access Controls & Break-Glass Protocol

- **Direct Access Disabled:** Direct kubectl exec and write permissions to production clusters are disabled for all individual engineers.
- **Emergency Break-Glass Access:** Requires dual approval via the #devops-emergency-access Slack workflow with mandatory automatic session termination after 2 hours.